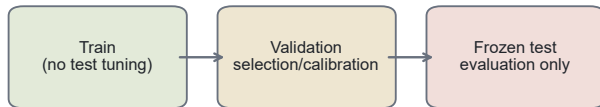
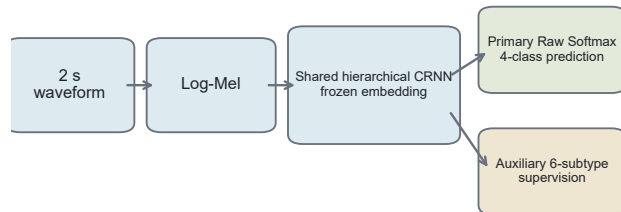
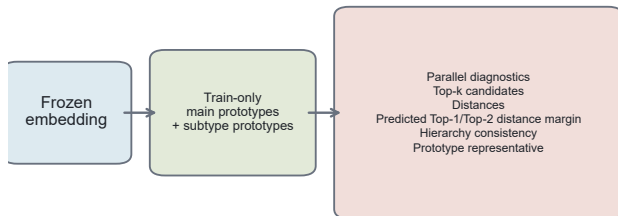
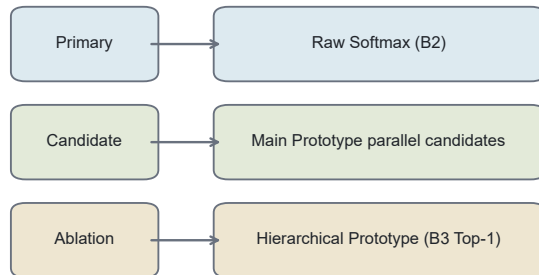


a Disjoint split roles

Exact path, source-ID and MD5 disjointness

b Validated representation and auxiliary supervision**c Parallel prototype candidate and diagnostic layer**

Validation-only calibration; frozen-test labels are evaluation only

d Evidence-bounded deployment interpretation

Parallel evaluation; no automatic routing claim